UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | INTRODUCTION TO INTERACTIVE MULTIMEDIA |
| COURSE CODE | CSC413 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 5 printed pages.*

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## PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

**1.** Keyframes define important positions in an animation.
**2.** Higher resolution always means a smaller file.
**3.** A plotter is an input device.
**4.** A storyboard is a visual plan of a multimedia sequence.
**5.** Anti-aliasing makes the edges of text look more jagged.
**6.** 3D animation does not use modelling.
**7.** A bitmap image is made up of pixels.
**8.** Authoring tools help developers organise and edit multimedia elements.
**9.** Interactivity gives users control over content and flow.
**10.** Interlacing displays every line of a video frame at the same time.
**11.** A linear presentation allows users to choose their own path.
**12.** Multimedia combines text, graphics, audio, video and animation.
**13.** A multimedia project always uses every media element.
**14.** A URL identifies the address of a web resource.
**15.** A prototype is an early model of a product.

## PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

**Table A**

| | | | | |
|---|---|---|---|---|
| A. Multimedia | B. Vector | C. MIDI | D. Storyboard | E. Plotter |
| F. Bitmap | G. Scanner | H. Cel animation | I. Serif | J. Browser |
| K. Keyframe | L. Sampling | M. Card-based tool | N. Tweening | O. Virtual reality |

**1.** image made of pixels
**2.** visual sequence plan
**3.** converting sound into digital values
**4.** important frame in an animation
**5.** image made of mathematical objects
**6.** input device that digitises documents
**7.** generating in-between frames
**8.** simulated environment
**9.** font with small strokes at character ends
**10.** standard for musical notes and instructions
**11.** authoring tool organised like pages or cards
**12.** software that displays web pages
**13.** integration of text, graphics, audio, video and animation
**14.** hand-drawn frame-by-frame animation
**15.** output device for large drawings

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## PART C (30 MARKS)

Choose the correct answer for each question.

**1.** Interactivity in multimedia means:
A. There is no navigation
B. Only text is shown
C. The system plays automatically
D. The user can control content and flow

**2.** Streaming allows media to:
A. Play while downloading
B. Only play offline
C. Require full download first
D. Remove compression

**3.** Chroma key is used to:
A. Add text
B. Scan images
C. Replace a colour background
D. Compress audio

**4.** Keyframes are:
A. Colour profiles
B. Audio samples
C. Important frames in an animation
D. Web links

**5.** Which is an audio file format?
A. WAV
B. JPEG
C. MPEG
D. HTML

**6.** HTML is used to:
A. Structure web content
B. Compress video
C. Store audio samples
D. Model 3D objects

**7.** MPEG is:
A. An image editor
B. A video compression family
C. A font
D. A storage device

**8.** Hypermedia is best described as:
A. A printed manual
B. A single video file
C. Text-only media
D. Multimedia with hyperlinks

**9.** Card-based authoring tools are suited to:
A. Live video only
B. 3D modelling only
C. Audio editing only
D. Messages with a beginning and end

**10.** A graphics tablet is used for:
A. Web browsing
B. Drawing input
C. Printing
D. Sound output

**11.** Frame rate is:
A. Frames per second
B. Pixels per inch
C. Samples per second
D. Bits per second

**12.** A wireframe is:
A. A codec
B. A final video
C. A low-fidelity layout sketch
D. A sound file

**13.** Interlacing:
A. Adds sound
B. Removes colour
C. Draws every line at once
D. Draws alternate lines of a frame

**14.** Anti-aliasing is used to:
A. Increase file size only
B. Add sound
C. Compress video
D. Smooth jagged edges

**15.** CMYK is used mainly for:
A. Printed output
B. Screen display
C. Animation
D. Sound mixing

**16.** A higher sampling rate generally:
A. Removes audio
B. Changes colour
C. Improves accuracy
D. Reduces accuracy

**17.** A bitmap image is made of:

A. Mathematical objects
B. Text characters
C. Sound samples
D. Pixels

**18.** A browser is used to:

A. Display web pages
B. Edit audio
C. Print documents only
D. Scan images

**19.** A presentation where the user controls the sequence is called:

A. Printed
B. Static
C. Linear
D. Non-linear

**20.** Kerning adjusts:

A. Space between individual character pairs
B. Font colour
C. Page size
D. Line length

**21.** Convergence in multimedia refers to:

A. Deleting media files
B. Merging of media and computing
C. Printing images
D. Recording audio only

**22.** A licence gives:

A. Ownership of all works
B. Permission to use a work
C. Free hardware
D. Unlimited bandwidth

**23.** RAM is:

A. Temporary working memory
B. An input device
C. Permanent storage
D. A web browser

**24.** A prototype is:

A. A codec
B. A font
C. A final release
D. An early model of a product

**25.** Which is an input device?

A. Printer
B. Scanner
C. Monitor
D. Plotter

**26.** Cel animation is created by:

A. Drawing each frame by hand
B. Printing images
C. Recording sound
D. Scanning text

**27.** Which is an example of linear multimedia?

A. A cinema presentation
B. A website with menus
C. A game with choices
D. An interactive kiosk

**28.** Sampling converts:

A. Digital video to text
B. Text to speech
C. Images to vectors
D. Analog sound to digital values

**29.** Amplitude of a sound wave relates to:

A. Loudness
B. Pitch
C. Frame rate
D. Colour

**30.** Resolution refers to:

A. Font size
B. Colour name
C. Number of pixels in an image
D. File type

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## PART D (40 MARKS)

### QUESTION 1

A company wants an interactive product demonstration that runs on a kiosk and on the web.

**a)** State TWO features to consider when choosing authoring software.

*(3 marks)*

**b)** Explain the interactivity level suitable for the demonstration.

*(3 marks)*

**c)** Explain TWO authoring tool categories appropriate for this project.

*(4 marks)*

### QUESTION 2

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

**a)** State TWO animation file formats suitable for this project.

*(3 marks)*

**b)** Describe the steps of cel animation and ONE advantage of using it.

*(3 marks)*

**c)** Differentiate 2-D and 2.5-D animation with ONE example for each.

*(4 marks)*

### QUESTION 3

A product catalogue must show sharp images on screen and in print.

**a)** State TWO differences between bitmap and vector images.

*(2 marks)*

**b)** Describe TWO factors that affect image quality.

*(4 marks)*

**c)** Explain how colour model and resolution affect screen and print output.

*(4 marks)*

### QUESTION 4

A museum wants a website that presents its collection and lets visitors explore related exhibits.

**a)** State TWO differences between linear and non-linear navigation.

*(2 marks)*

**b)** Describe TWO roles of hyperlinks in the website.

*(3 marks)*

**c)** Explain how streaming and compression affect multimedia delivery on the web.

*(5 marks)*

*END OF QUESTION PAPER*